Neeli Praveen

Mangalagiri, Andhra Pradesh, India | praveen_neeli@srmap.edu.in | +91-93928-01318
Portfolio | linkedin.com/in/neelipraveen | github.com/neelipraveen

Research Experience

Research Intern — **Mar 2026 – Jun 2026**
SRM University AP – Research Internship — *Andhra Pradesh, India*

- Designed and trained a federated deep learning pipeline (MQTT-Fed InceptionTime) for cross-dataset ECG arrhythmia classification, combining a 0.47M-parameter time-series model with FedProx aggregation across 5 simulated clients and Dirichlet non-IID partitioning to reflect realistic cross-hospital data variation
- Added guarded rehearsal fine-tuning to retain performance across hospitals and shipped the model through a full-stack web application (FastAPI + React, ONNX runtime inference), reaching a final balanced F1 > 0.93.

Projects

CERVICAL SPINE FRACTURE DETECTION — Deep Learning

- Engineered a segmentation-guided deep learning framework to detect cervical spine fractures in 3D CT scans, benchmarking 8 architectures including MViT and Swin Transformers for diagnostic precision
- Tackled extreme class imbalance in the RSNA 2022 medical dataset using Weighted Binary Cross-Entropy and Focal Loss, improving model reliability and detection sensitivity

STABILIZER V2: HEART DISEASE PREDICTION — Deep Learning

- Designed the Stabilizer V2 protocol combining Transformers with Gated Residual Networks (GRN) and TVAE-based data augmentation, scaling the training corpus to 5,000 records
- Applied Monte Carlo Dropout for uncertainty quantification and SHAP for model interpretability aligned with clinical markers such as chest pain type

DISTRIBUTED ANOMALY DETECTION SYSTEM — Big Data & Deep Learning

- Built a high-throughput pipeline using Apache Flume, HDFS, and PySpark to concurrently process 67 data streams; architected an attention-augmented model (CNN + BiLSTM + Self-Attention) for industrial sensor anomaly detection
- Implemented a leakage-free dynamic thresholding strategy with Exponential Weighted Moving Averages (EWMA), achieving F1 $\geq$ 0.6 on approximately 50% of test datasets with automated retraining and HDFS artifact storage

Education

SRM University Andhra Pradesh — **Aug 2025 – Ongoing**
Master of Technology in Data Science — *CGPA: 9.38*

National Institute of Technology Andhra Pradesh — **Jun 2024**
Bachelor of Technology in Computer Science & Engineering — *First Class*

Skills

Programming Languages: Python, SQL, HTML/CSS
Big Data & Cloud: Apache Spark, Apache Kafka, Hadoop (HDFS, YARN, MapReduce), Apache Flume
Machine Learning: Deep Learning (Transformers, CNNs), NLP, PyTorch, TensorFlow, Scikit-learn, SHAP
Libraries & Tools: Pandas, NumPy, Matplotlib, Git, Jupyter Notebook, VS Code, Web Scraping

Publications & Certifications

- Published **"Evaluating CNNs and Transformers for Multi-Label Classification in 3D CT Imaging"** at the 6th International Conference on Recent Advances in Information Technology (RAIT 2025), IEEE – May 2025
- Completed the **NPTEL – Applied Accelerated Artificial Intelligence** certification offered by IIT Madras, achieving Silver grade – Oct 2023